

D211: Advanced Data Acquisition

Hannah R. Bergman

WGU Email: hbergm3@wgu.edu

Table of Contents

Part 1: Data Dashboard..... 2

 A. Tableau Dashboard..... 2

 A1. Datasets & Data Sources..... 2

 A2. Dashboard Access Instructions..... 2

 A3. Story & Dashboard Navigation..... 3

 A4. SQL Code Supporting the Dashboard..... 7

Part 2: Demonstration..... 12

 B. Panopto Presentation..... 12

Part 3: Report..... 13

 C. Exploration of the Data Using Advanced SQL Operations..... 13

 C1. Dashboard Purpose & Function..... 13

 C2. Business Intelligence Tool Justification..... 13

 C3. Data Cleaning & Preparation..... 13

 C4. Dashboard Creation Summary..... 14

 C5. Data Analysis Results..... 22

 C6. Limitations of the Analysis..... 24

 D. Data & Code Sources..... 24

 E. References..... 25

Part 1: Data Dashboard

A. Tableau Dashboard

The dashboard is attached as the file:
hrbergman_d211_task1.twbx

A1. Datasets & Data Sources

Our analysis makes use of two datasets:

1. Provided Internal Data: churn database

The internal data used in this analysis consists of the following churn tables from within the PostgreSQL database provided by Western Governors University: customer, contract, job, location, and payment. This dataset comprises 10,000 customer records from within our organization, reflecting our specific customer population (WGU, 2023).

2. External Dataset: telco_customer_churn_dataset_full.csv

The second dataset, 'telco_customer_churn_dataset_full.csv', was acquired from [Kaggle](#). This dataset contains information about a competing telecommunications company, Telco, which provided home phone and internet services to 7,043 customers (IBM, 2019). This dataset aligns with our own in terms of the factors used to measure customer churn, facilitating a direct comparison.

A2. Dashboard Access Instructions

The dashboard is provided as a .twbx file, which can be viewed using the Tableau 2021.4 desktop application that is pre-installed on the Labs on Demand environment. You can view the dashboard through Labs on Demand (LoD) by following these step-by-step instructions:

1. Log into the LoD environment using your user credentials.
2. Access the files 'hrbergman_d211_task1.twbx', 'hrbergman_d211_report.pdf', and 'telco_customer_churn_dataset_full.csv' which should all be attached to this report submission.
 - a. Save these files to the Downloads folder within the virtual machine (VM) environment.
3. On the VM desktop, double-click on the pgAdmin application to access the Churn database:
 - a. Once inside pgAdmin, right click beside the "churn" database and select the Query Tool
 - b. Copy, paste and execute the SQL scripts from [Section A4, Steps 1 & 2](#), of this report (you should be able to access this SQL code within the VM via the 'hrbergman_d211_report.pdf' file you downloaded into the VM environment). Running these scripts will create a new table within the database called 'telco'.
 - c. [Section A4, Step 3](#) details how to populate this newly created table with data from the 'telco_customer_churn_dataset_full.csv' file.
 - d. Execute the SQL scripts under [Section A4, Step 4](#) to clean this dataset and standardize the data within both the telco and customer tables within the churn database.

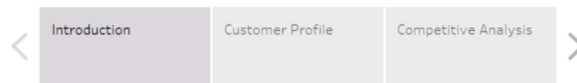
- e. Lastly, execute the SQL scripts under [Section A4. Step 5](#) to create another table that combines data from both the telco and wgu tables, which will act as the data source for the 'hrbergman_d211_task1.twbx' Tableau file.
 - f. When all of the required SQL scripts have been executed, minimize or exit out of pgAdmin and return to the VM desktop.
4. Since Tableau is already installed on the VM, double-click on the Tableau 2021.4 Desktop icon to open this application.
 - a. At the top left, click the File tab then select 'Open...'. Navigate to the location of the downloaded 'hrbergman_d211_task1.twbx' file, select it and click the Open button to open the file.
 - b. When prompted to sign in to complete connection to the data source, enter the following credentials and then click the Sign In button:
 - i. Username: postgres
 - ii. Password: Passw0rd!
 5. The dashboard should now appear and be fully operational

A3. Story & Dashboard Navigation

Introduction

The first story point is a title page that serves as a brief introduction to the Tableau story and the presenter, in a similar format to a PowerPoint presentation or slide deck.

TELECOMMUNICATIONS ANALYSIS & COMPARISON



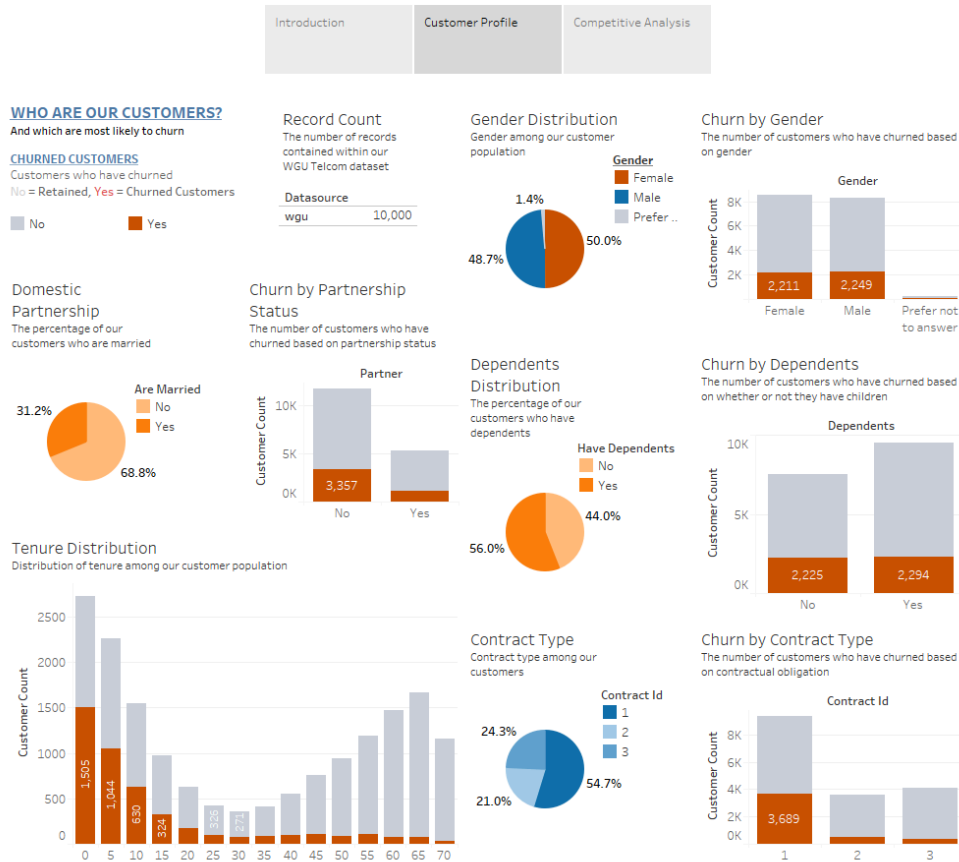
WGU TELECOMMUNICATIONS

A comparative Analysis of WGU and Telco
Customer Demographics & Behaviors

Customer Profile

The second story point contains the Customer Profile Dashboard, a detailed visualization of our customer base's demographics and churn rates.

TELECOMMUNICATIONS ANALYSIS & COMPARISON



Customer Profile Dashboard Features:

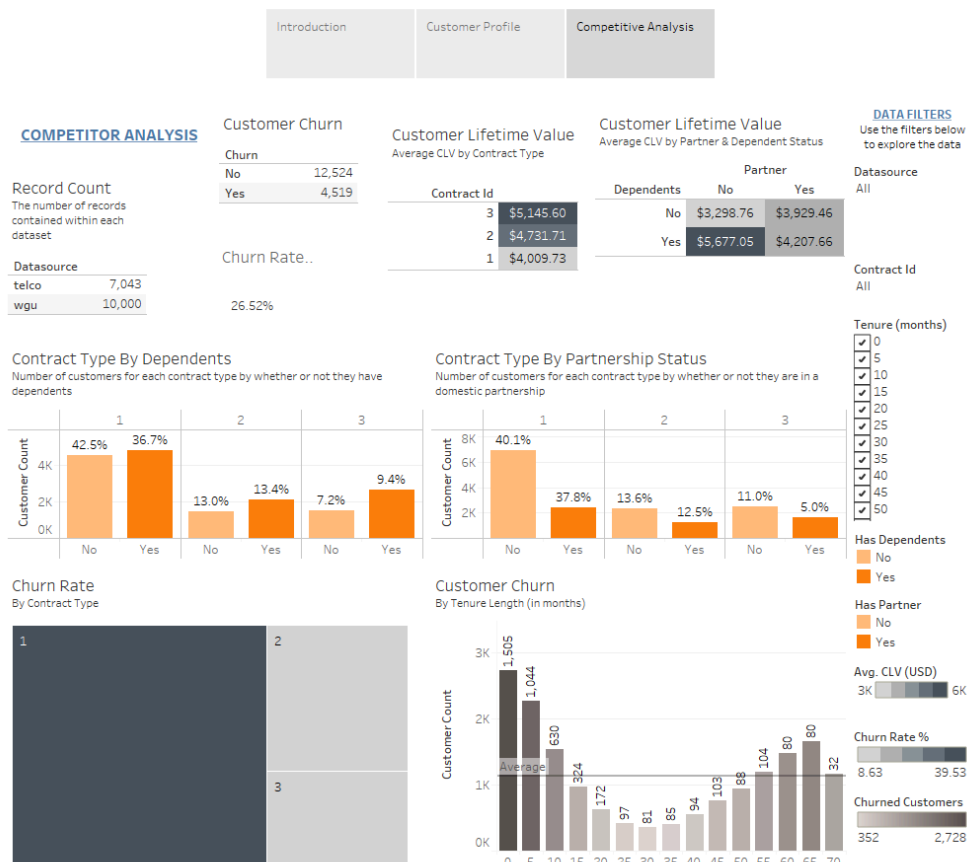
- **Churn Filter:** Under 'CHURNED CUSTOMERS' is a filter that allows users to toggle between churned and retained customers.
- **Record Count:** Displays the total number of records in the WGU churn dataset.
- **Gender Distribution:** A pie chart showing the gender distribution among our customer population. This also shows a legend that acts as a filter that allows the user to toggle between Female, Male, and Nonbinary.
- **Churn by Gender:** A stacked bar chart highlighting the number of churned customers by gender.
- **Domestic Partnership:** A pie chart showing the distribution of domestic partnership status (married vs. single). This visual also includes a legend acting as a filter that allows the user to toggle between those who are married (Yes) and those who aren't (No).
- **Churn by Partnership Status:** A stacked bar chart displaying the number of churned customers by partnership status.
- **Dependents Distribution:** A pie chart showing the distribution of customers with and without dependents. This also has a legend that doubles as a filter to toggle between those who have children (Yes) and those who don't (No).

- **Churn by Dependents:** A stacked bar chart highlighting the number of churned customers by dependency status.
- **Tenure Distribution:** A histogram showing the distribution of customer tenure.
- **Contract Type:** A pie chart illustrating the distribution of contract types. Again, the legend for this chart is also a filter that allows the user to toggle between 1 (Month-to-month), 2 (Two year), and 3 (One year) contract type.
- **Churn by Contract Type:** A stacked bar chart showing the number of churned customers by contract type.

Competitor Analysis

The third story point contains the Competitor Analysis Dashboard, which makes use of both the internal and external datasets to show information about our company and our competitor.

TELECOMMUNICATIONS ANALYSIS & COMPARISON



Competitor Analysis Dashboard Features:

- **Record Count:** Displays the total number of records in the WGU churn dataset and/or Telco Dataset(s).
- **Customer churn:** Shows the total count of customers who did and did not churn in either or both datasets.

- **Churn Rate:** Displays the individual or combined Churn Rate percentage which was manually calculated within this worksheet as the number of churned customers divided by the total number of customers multiplied by 100 to get this percentage.
- **Customer Lifetime Value (by Contract Type):** Shows the average Customer Lifetime Value (CLV) by contract type. These numbers were manually calculated within the worksheet related to this highlight table, by multiplying tenure by monthly charges and taking the average of the totals to get the average lifetime value of each customer by contract type.
- **Customer Lifetime Value (by Marital & Dependent Status):** Shows the average Customer Lifetime Value (CLV) by partner and dependent status. These numbers were manually calculated within the worksheet related to this highlight table, by multiplying tenure by monthly charges and taking the average of the totals to get the average lifetime value of each customer by dependent and partnership status.
- **Contract Type By Dependents:** This side-by-side bar chart shows the total number of customers (Customer Count) for each contract type by whether or not they have dependents (Yes/No) and displays a comparative churn rate percentage as a label above each bar.
- **Contract Type By Partnership Status:** This side-by-side bar chart shows the total number of customers (Customer Count) for each contract type by whether or not they are in a domestic partnership (Yes/No) and displays a comparative churn rate percentage as a label above each bar.
- **Average Tenure Length:** This treemap displays the average tenure length by contract type.
- **Customer Churn:** This bar chart provides the number of customers who churned by their average tenure length. A reference line has been added to show the average churn across all tenures.
- **Data Filters:** This column provides various filters to better explore the data provided within this dashboard
 - **Change Dataset:** This interactive radio button list allows the user to change datasets to view All, only the Telco dataset, or only the WGU Telco dataset
 - **Contract Type:** This dropdown allows the user to select a single, multiple or all contract types to filter data.
 - **Tenure (months):** This checklist filters the Customer Churn visual, allowing the user to select all or individual tenure length (by number of months).
 - **Has Dependents:** This is a legend for the Contract Type By Children that also acts as a filter for this visual and the CLV by Partner & Dependent Status highlight table.
 - **Has Partner:** This is a legend for the Contract Type By Partnership Status that also acts as a filter for this visual and the CLV by Partner & Dependent Status highlight table.
 - **Avg. Tenure (months):** This legend explains the meaning of the colors used in the Average Tenure Length treemap.
 - **Avg. CLV (USD):** This legend explains the meaning of the colors used in the Customer Lifetime Value highlight tables.

- **Churned Customers:** This legend explains the meaning of the colors used in the Customer Churn graph.

A4. SQL Code Supporting the Dashboard

/* Step 1. Locate or create the base customer table with the supplied internal data */

-- SQL to reconstruct the customer table within the churn database, if needed

CREATE TABLE public.customer

(

customer_id text COLLATE pg_catalog."default" NOT NULL,

lat numeric,

lng numeric,

population integer,

children integer,

age integer,

income numeric,

marital text COLLATE pg_catalog."default",

churn text COLLATE pg_catalog."default",

gender text COLLATE pg_catalog."default",

tenure numeric,

monthly_charge numeric,

bandwidth_gp_year numeric,

outage_sec_week numeric,

email integer,

contacts integer,

yearly_equip_faiure integer,

techie text COLLATE pg_catalog."default",

port_modem text COLLATE pg_catalog."default",

tablet text COLLATE pg_catalog."default",

job_id integer,

payment_id integer,

contract_id integer,

location_id integer,

CONSTRAINT customer_pkey PRIMARY KEY (customer_id),

CONSTRAINT customer_contract_id_fkey FOREIGN KEY (contract_id)

REFERENCES public.contract (contract_id) MATCH SIMPLE

ON UPDATE NO ACTION

ON DELETE NO ACTION

NOT VALID,

CONSTRAINT customer_job_id_fkey FOREIGN KEY (job_id)

REFERENCES public.job (job_id) MATCH SIMPLE

ON UPDATE NO ACTION

ON DELETE NO ACTION

NOT VALID,

```

CONSTRAINT customer_location_id_fkey FOREIGN KEY (location_id)
REFERENCES public.location (location_id) MATCH SIMPLE
ON UPDATE NO ACTION
ON DELETE NO ACTION
NOT VALID,
CONSTRAINT customer_payment_id_fkey FOREIGN KEY (payment_id)
REFERENCES public.payment (payment_id) MATCH SIMPLE
ON UPDATE NO ACTION
ON DELETE NO ACTION
NOT VALID
)

```

```

TABLESPACE pg_default;

```

```

ALTER TABLE public.customer
OWNER to postgres;

```

/* Step 2. Create a new telco table to store the telecommunications data from the external Telco dataset */

-- SQL to create the telco table with the required columns

```

CREATE TABLE public.telco (
customer_id TEXT PRIMARY KEY,
gender TEXT,
senior_citizen INTEGER,
partner TEXT,
dependents TEXT,
tenure INTEGER,
phone_service TEXT,
multiple_lines TEXT,
internet_service TEXT,
online_security TEXT,
online_backup TEXT,
device_protection TEXT,
tech_support TEXT,
streaming_tv TEXT,
streaming_movies TEXT,
contract TEXT,
paperless_billing TEXT,
payment_method TEXT,
monthly_charges NUMERIC,
churn TEXT
);

```

/* Step 3. Populate the newly created table with data from the downloaded csv */

Right click on the 'telco' table within the database > click Query Tool > enter into the Query Editor and perform the following steps:

1. Toggle Import/Export to 'Import'
2. Select the external csv file, 'telco_customer_churn_dataset_full.csv', from the VM's Downloads folder
3. Make sure the format is set to csv
4. From the dropdown list under Delimiter, select the comma ",",
5. Click "OK"
6. Toggle Header to "Yes" to indicate the first row is the header row

/* Step 4. Clean and standardize the data within the tables */

-- Update contract types to match the id's used in the customer table

```
UPDATE telco
SET "contract" =
CASE
  WHEN "contract" = 'Month-to-month' THEN '1'
  WHEN "contract" IN ('1 year', 'One year') THEN '2'
  WHEN "contract" = 'Two year' THEN '3'
  ELSE "contract"
END;
```

-- Rename contract to contract_id

```
ALTER TABLE telco
RENAME COLUMN contract TO contract_id;
```

-- Change data type of contract_id from text to int

```
ALTER TABLE telco
ALTER COLUMN contract_id TYPE INTEGER
USING contract_id::INTEGER;
```

-- Update payment methods to match the id's used in the customer table

```
UPDATE telco
SET "payment_method" =
CASE
  WHEN "payment_method" = 'Bank transfer (automatic)' THEN '1'
  WHEN "payment_method" = 'Credit card (automatic)' THEN '2'
  WHEN "payment_method" = 'Electronic check' THEN '3'
  WHEN "payment_method" = 'Mailed check' THEN '4'
  ELSE "payment_method"
END;
```

-- Rename payment_method to payment_id

```
ALTER TABLE telco
```

```

RENAME COLUMN payment_method TO payment_id;

-- Change data type of payment_id from text to int
ALTER TABLE telco
ALTER COLUMN payment_id TYPE INTEGER
USING payment_id::INTEGER;

-- Rename age to senior_citizen
ALTER TABLE customer
RENAME COLUMN age TO senior_citizen;

-- Update senior citizen column - if age is >= 65 they're a senior otherwise not
UPDATE customer
SET senior_citizen =
CASE
    WHEN senior_citizen >= 65 THEN 1
    ELSE 0
END;

-- Change data type of tenure from integer to numeric
ALTER TABLE telco
ALTER COLUMN tenure TYPE NUMERIC
USING tenure::NUMERIC;

-- Round down precision of tenure in customer table
UPDATE customer
SET tenure = ROUND(tenure, 2);

-- Round down precision of monthly_charge in customer table
UPDATE customer
SET monthly_charge = ROUND(monthly_charge, 2);

-- Rename monthly_charges to monthly_charge
ALTER TABLE telco
RENAME COLUMN monthly_charges TO monthly_charge;

-- Add a new column called datasource with the value wgu to all rows in the customer table
ALTER TABLE customer
ADD COLUMN datasource TEXT DEFAULT 'wgu';

-- Add a new column called datasource with the value telco to all rows in the telco table
ALTER TABLE telco
ADD COLUMN datasource TEXT DEFAULT 'telco';

```

```

-- Rename children to dependents
ALTER TABLE customer
RENAME COLUMN children TO dependents;

-- Change data type of dependents from integer to text
ALTER TABLE customer
ALTER COLUMN dependents TYPE TEXT USING dependents::TEXT;
-- Update dependents column - if number of children is >= 1 they have dependents and the
value is yes otherwise the value is no
UPDATE customer
SET dependents =
    CASE
        WHEN dependents::INTEGER >= 1 THEN 'Yes'
        ELSE 'No'
    END;

-- Rename marital to partner
ALTER TABLE customer
RENAME COLUMN marital TO partner;

-- Update partner column - if value is married then the individual has a partner otherwise the
value is no
UPDATE customer
SET partner =
    CASE
        WHEN partner = 'Married' THEN 'Yes'
        ELSE 'No'
    END;

/* Step 5. Create a 'combined' table that consolidates the specified columns from both
tables */
-- SQL to create the combined table
CREATE TABLE combined (
    customer_id TEXT,
    gender TEXT,
    churn TEXT,
    tenure NUMERIC,
    monthly_charge NUMERIC,
    partner TEXT,
    dependents TEXT,
    contract_id INTEGER,
    payment_id INTEGER,
    datasource TEXT
);

```

```
-- Insert data from both tables into the new table combining the data using UNION ALL
INSERT INTO combined (customer_id, gender, churn, tenure, monthly_charge, partner,
dependents, contract_id, payment_id, datasource)
SELECT
    customer_id,
    gender,
    churn,
    CAST(tenure AS NUMERIC) AS tenure,
    CAST(monthly_charge AS NUMERIC) AS monthly_charge,
    partner,
    dependents,
    contract_id,
    payment_id,
    'telco' AS datasource
FROM telco
UNION ALL
SELECT
    customer_id,
    gender,
    churn,
    CAST(tenure AS NUMERIC) AS tenure,
    CAST(monthly_charge AS NUMERIC) AS monthly_charge,
    partner,
    dependents,
    contract_id,
    payment_id,
    'customer' AS datasource
FROM customer;
```

```
-- Update the datasource from customer to wgu in the combined table
UPDATE combined
SET datasource = 'wgu'
WHERE datasource = 'customer';
```

Part 2: Demonstration

B. Panopto Presentation

Panopto video presentation:

<https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=d09b4d64-932a-4065-8291-b2450129a327>

Part 3: Report

C. Exploration of the Data Using Advanced SQL Operations

C1. Dashboard Purpose & Function

In today's competitive telecommunications market, customers have numerous service provider options. While this certainly benefits consumers, it also presents a significant challenge for companies like ours. High churn rates, reaching up to 25% annually, are prevalent in this industry. Considering the cost of acquiring new customers compared to retaining existing ones, customer retention is crucial. Our dashboard and research aims to identify customer characteristics and predict churn risk to enhance our retention strategies. With our dashboard, we can analyze churn data from our company and a key competitor, to help refine our strategies to mitigate churn risk and increase customer tenure.

The supplemental competitor dataset's variables were carefully chosen due to their similarity to our own data, providing valuable insights into our own customer base. By comparing our data with the competitor data, we can gain a deeper understanding of customer demographics and churn factors.

C2. Business Intelligence Tool Justification

The business intelligence tool employed for this analysis is Tableau, a powerful visual analytics tool. Tableau empowers users to visualize, understand, and act upon data through the creation of intricate yet intuitive and interactive data visualizations. The platform's user-friendly interface facilitates the effective communication of our analysis findings to stakeholders, regardless of their technical expertise. Additionally, Tableau's robust capabilities enable seamless exploration of complex datasets to uncover hidden patterns and meaningful insights.

C3. Data Cleaning & Preparation

To ensure data quality and consistency, the datasets underwent the following data cleaning and preparation steps, which I translated from python to SQL from my last project (Bergman, 2024):

1. Column Renaming

To adhere to standard, pythonic naming conventions (maintaining lowercase and separating words by underscores), I updated column labels within both the customer and telco tables to ensure consistency and readability.

- Telco Table:
 - **customerID** to **customer_id**
 - **SeniorCitizen** to **senior_citizen**
 - **Contract** to **contract_id**
 - **PaymentMethod** to **payment_id**
 - **MonthlyCharges** to **monthly_charge**

- Customer Table:
 - **age** to **senior_citizen** (to match the Telco dataset)
 - **marital** to **partner** (to match the Telco dataset)
 - **children** to **dependents** (to match the Telco dataset)

2. Value Adjustments

- **Contract** → **contract_id**: I converted the contract types in the Telco dataset from text to integer format. This was done by assigning each contract type a unique integer ID based on the information in the Customer and Contract tables.
- **PaymentMethod** → **payment_id**: Within the Telco dataset, I updated the payment method types to match the corresponding ID numbers associated with each within the Customer and Payment tables. In addition to this, I also changed the data type from text to integer to reflect this numerical change.
- **Age** → **senior_citizen**: Within the Customer dataset, I updated age to reflect senior status. When values are greater than or equal to 65, the age value is relabeled as 1 to indicate the individual is a senior citizen or 0 to indicate they are not.
- **tenure**: Within the Telco table, tenure was changed from an integer to a numeric data type. Additionally, this value was rounded down in precision within the Customer table.
- **monthly_charge**: This value was rounded down by 2 to avoid excessive precision in the Customer table.
- **Marital** → **partner**: Within the marital status column of the Customer table, we have values indicating whether customers are 'Divorced', 'Widowed', 'Never Married', 'Separated' or 'Married'. Since the telco dataset only measures whether someone is in a domestic partnership or not, I chose to remap these values with 'Married' remapped to Yes to indicate domestic partnership and the other values remapped to No.
- **Children** → **dependents**: The Customer table records the exact number of children in a customer's household instead of the number of dependents. To account for this, I created a new 'dependents' column for the Customer table which marks a value of Yes if a customer has more than 1 child and No for childfree and childless homes.
- **datasource**: To differentiate between the two combined tables, a new column named 'datasource' was added to each dataset to identify the source.

C4. Dashboard Creation Summary

Sheet 1: Domestic Partnership (Pie Chart)

- Double click on the tab that says 'Sheet 1' or right click on the tab and select 'Rename' and rename the worksheet to '**Partner**'
- Drag the '**Partner**' Dimension to the Color mark
- Drag the '**combined (Count)**' Measure to the Size mark
- Click the 'Show Me' tab > select the Pie Chart
- Drag the '**combined (Count)**' Measure to the Label mark
 - Under this mark, click the CNT(combined) dropdown > select Quick Table Calculation > select 'Percent of Total'
- Drag the '**combined (Count)**' Measure to the Tooltip mark
- Double-click Tooltip in the Marks column

- Add the words 'Partnership Distribution' to the top, change the font to Tableau Book, font size 10, make bold, underline and change color to solid black
- Edit '**Partner**' to say '**Customer is married or in a domestic partnership**', adjust spacing
- Edit 'Count of combined' to say 'Number of customers', adjust spacing
- Edit '% of Total Count of combined along Table (Across)' to say '% of all Customers', adjust spacing
- Click OK to save changes
- Drag the '**Datasource**' Dimension to the Filters column
- Select 'All' or check both the 'telco' and 'wgu' boxes under the General tab > click OK
- You may see 2 filter boxes to the right of the graph at this point, one that says 'Partner' and another that says 'CNT(combined)'
- Click on the dropdown arrow on the 'CNT(combined)' box and select 'Hide Card' since we won't be using this to filter this data
- Double click on the 'Partner' title at the top of the graph, edit the title to
- 'Domestic Partnership' hit return 'The percentage of our customers who are married'
- Select 'Domestic Partnership' text and edit to Tableau Light, font size 12 to make this the main title
- Select 'The percentage of our customers who are married' text and edit to Tableau Light, font size 8 to create a subheading
- Click the dropdown of the 'Partner' filter box > select Edit Title... > change the Legend Title to 'Are Married' > click OK
- Click the dropdown of the 'Partner' filter box > select Edit Colors.. > from the 'Select Color Palette:' dropdown, select the 'Color Blind' palette > under 'Select Data Item:' click 'No' and assign this item to the lightest orange color of this color palette, click 'Yes' and assign this time to the darkest orange color > click 'OK'

Sheet 2: Churn by Partnership Status (Stacked Bar Chart)

- Click the 'New Worksheet' tab > change the name of the sheet from Sheet 2 to '**Partner Ch.**'
- Drag the '**Partner**' Dimension to Columns
- Drag '**CNT(combined)**' Measure to Rows
- Drag '**Churn**' to Color under Marks
- Under the Analysis tab > select 'Create Calculated Field..' > name this **Churned Customers** > enter the formula IF [Churn] = "Yes" THEN 1 ELSE 0 END > click 'OK'
- Drag this new '**Churned Customers**' Measure to Tooltip under Marks
- Under the Analysis tab > select 'Create Calculated Field..' > name this **Retained Customers** > enter the formula IF [Churn] = "No" THEN 1 ELSE 0 END > click 'OK'
- Drag this new '**Retained Customers**' Measure to Tooltip under Marks
- Drag the '**Churn**' Dimension to Label under Marks
- Click the dropdown under the Churn Label > select 'Measure' > select 'Count'
- Drag the '**Datasource**' Dimension to the Filters column > click 'All' > click 'OK'
- Double click Tooltip under the Marks column:
 - Add a title of 'Churn by Partnership Status', bold and underline this text
 - Update 'Partner' text to 'Customer is in domestic partnership'

- Remove the CNT(combined) line
- Remove the CNT(Churn) line
- Change 'Churned Customers' to 'Number of Churned Customers', make this text red
- Change 'Retained Customers' to 'Number of Retained Customers', make this text green
- Add the text *Yes = Customers who DID churn, *No = Customers who DID NOT churn to the bottom of the Tooltip to add further clarification for stakeholders reviewing this data, adjust size of this text to 8
- Click 'OK' to save
- Double click the 'Partner Ch.' title and edit this to say 'Churn by Partnership Status, The number of customers who have churned based on partnership status' using the same formatting as in Sheet 1 (font sizes 12 & 8)
- Click the 'Churn' filter box and select 'Edit Colors...' from the dropdown > select the Color Blind color palette > make 'No' the lightest grey shade and 'Yes' the darkest red shade > click 'OK'
- Since we only really care about churned customers in our analysis, right click on the now grey portion of each bar of the now stacked bar chart > select 'Mark Label' > select 'Never Show' to hide the totals of the retained customers
- Double-click the axis label that says 'Count of combined' > change the Axis Title to 'Customer Count' and then click 'X' to close this dialog box

Sheet 3: Dependents Distribution (Pie Chart)

- Double click on the tab that says 'Sheet 3' or right click on the tab and select 'Rename' and rename the worksheet to '**Dependents**'
- Drag the '**Dependents**' Dimension to the Color mark
- Drag the '**combined (Count)**' Measure to the Size mark
- Click the 'Show Me' tab > select the Pie Chart
- Drag the '**combined (Count)**' Measure to the Label mark
- Under this mark, click the CNT(combined) dropdown > select Quick Table Calculation > select 'Percent of Total'
- Drag the '**combined (Count)**' Measure to the Tooltip mark
- Double-click Tooltip in the Marks column and make appropriate edits for the data contained within this graph
- Drag the '**Datasource**' Dimension to the Filters column. Select 'All' or check both the 'telco' and 'wgu' boxes under the General tab > click OK
- Double click on the title at the top of the graph and edit
- Click the dropdown of the filter box and select Edit Title... to edit the title
- Click the dropdown of the filter box > select Edit Colors.. > from the 'Select Color Palette:' dropdown, select the 'Color Blind' palette > under 'Select Data Item:' click 'No' and assign this item to the lightest orange color of this color palette, click 'Yes' and assign this time to the darkest orange color > click 'OK'

Sheet 4: Churn by Dependents (Stacked Bar Chart)

- Click the 'New Worksheet' tab > change the name of the sheet
- Drag the '**Dependents**' Dimension to Columns
- Drag '**CNT(combined)**' Measure to Rows
- Drag '**Churn**' to Color under Marks
- Drag the '**Churned Customers**' & '**Retained Customers**' Measures to Tooltip under Marks
- Drag the '**Churn**' Dimension to Label under Marks
- Click the dropdown under the Churn Label > select 'Measure' > select 'Count'
- Drag the '**Datasource**' Dimension to the Filters column > click 'All' > click 'OK'
- Double click Tooltip under the Marks column and edit the Tooltips as appropriate
- Double click the title and edit
- Click the 'Churn' filter box and select 'Edit Colors...' from the dropdown > select the Color Blind color palette > make 'No' the lightest grey shade and 'Yes' the darkest red shade > click 'OK'
- Since we only really care about churned customers in our analysis, right click on the now grey portion of each bar of the now stacked bar chart > select 'Mark Label' > select 'Never Show' to hide the totals of the retained customers
- Double-click the axis label that says 'Count of combined' > change the Axis Title to 'Customer Count' and then click 'X' to close this dialog box

Sheet 5: Gender Distribution (Pie Chart)

- Click the 'New Worksheet' tab > change the name of the sheet
- Drag the '**Gender**' Dimension to the Color mark
- Drag the '**combined (Count)**' Measure to the Size mark
- Click the 'Show Me' tab > select the Pie Chart
- Drag the '**combined (Count)**' Measure to the Label mark
- Under this mark, click the CNT(combined) dropdown > select Quick Table Calculation > select 'Percent of Total'
- Drag the '**combined (Count)**' Measure to the Tooltip mark
- Double-click Tooltip in the Marks column and make appropriate edits for the data contained within this graph
- Drag the '**Datasource**' Dimension to the Filters column. Select 'All' or check both the 'telco' and 'wgu' boxes under the General tab > click OK
- Double click on the title at the top of the graph and edit
- Click the dropdown of the filter box and select Edit Title... to edit the title
- Click the dropdown of the filter box > select Edit Colors.. > from the 'Select Color Palette:' dropdown, select the 'Color Blind' palette > under 'Select Data Item:' click 'Female' and assign this item to the darkest red color of this color palette, click 'Male' and assign this item to the darkest blue color, click 'Prefer not to answer' and assign this item to the lightest grey color > click 'OK'

Sheet 6: Churn by Gender (Stacked Bar Chart)

- Click the 'New Worksheet' tab > change the name of the sheet
- Drag the '**Gender**' Dimension to Columns

- Drag '**CNT(combined)**' Measure to Rows
- Drag '**Churn**' to Color under Marks
- Drag the '**Churned Customers**' & '**Retained Customers**' Measures to Tooltip under Marks
- Drag the '**Churn**' Dimension to Label under Marks
- Click the dropdown under the Churn Label > select 'Measure' > select 'Count'
- Drag the '**Datasource**' Dimension to the Filters column > click 'All' > click 'OK'
- Double click Tooltip under the Marks column and edit the Tooltips as appropriate
- Double click the title and edit
- Click the 'Churn' filter box and select 'Edit Colors...' from the dropdown > select the Color Blind color palette > make 'No' the lightest grey shade and 'Yes' the darkest red shade > click 'OK'
- Since we only really care about churned customers in our analysis, right click on the now grey portion of each bar of the now stacked bar chart > select 'Mark Label' > select 'Never Show' to hide the totals of the retained customers
- Double-click the axis label that says 'Count of combined' > change the Axis Title to 'Customer Count' and then click 'X' to close this dialog box

Sheet 7: Contract Type (Pie Chart)

- Click the 'New Worksheet' tab > change the name of the sheet
- Drag the '**Contract Id**' Dimension to the Color mark
- Drag the '**combined (Count)**' Measure to the Size mark
- Click the 'Show Me' tab > select the Pie Chart
- Drag the '**combined (Count)**' Measure to the Label mark
- Under this mark, click the CNT(combined) dropdown > select Quick Table Calculation > select 'Percent of Total'
- Drag the '**combined (Count)**' Measure to the Tooltip mark
- Double-click Tooltip in the Marks column and make appropriate edits for the data contained within this graph
- Drag the '**Datasource**' Dimension to the Filters column. Select 'All' or check both the 'telco' and 'wgu' boxes under the General tab > click OK
- Double click on the title at the top of the graph and edit
- Click the dropdown of the filter box and select Edit Title... to edit the title
- Click the dropdown of the filter box > select Edit Colors.. > from the 'Select Color Palette:' dropdown, select the 'Color Blind' palette > under 'Select Data Item:' click '1' and assign this item to the darkest blue color of this color palette, click '2' and assign this item to the lightest blue color, click '3' and assign this item to the medium blue color > click 'OK'

Sheet 8: Churn by Contract Type (Stacked Bar Chart)

- Click the 'New Worksheet' tab > change the name of the sheet
- Drag the '**Contract Id**' Dimension to Columns
- Drag '**CNT(combined)**' Measure to Rows
- Drag '**Churn**' to Color under Marks
- Drag the '**Churned Customers**' & '**Retained Customers**' Measures to Tooltip under Marks
- Drag the '**Churn**' Dimension to Label under Marks

- Click the dropdown under the Churn Label > select 'Measure' > select 'Count'
- Drag the '**Datasource**' Dimension to the Filters column > click 'All' > click 'OK'
- Double click Tooltip under the Marks column and edit the Tooltips as appropriate
- Double click the title and edit
- Click the 'Churn' filter box and select 'Edit Colors...' from the dropdown > select the Color Blind color palette > make 'No' the lightest grey shade and 'Yes' the darkest red shade > click 'OK'
- Since we only really care about churned customers in our analysis, right click on the now grey portion of each bar of the now stacked bar chart > select 'Mark Label' > select 'Never Show' to hide the totals of the retained customers
- Double-click the axis label that says 'Count of combined' > change the Axis Title to 'Customer Count' and then click 'X' to close this dialog box

Sheet 9: Tenure Distribution (Bar Chart)

- Click the 'New Worksheet' tab > change the name of the sheet
- Right click the 'Tenure' Measure > Create > Bins... > name this 'Tenure (bins)', set the Size of bins to 5 > click OK
- Drag the new '**Tenure (bins)**' Dimension to Columns
- Drag '**CNT(combined)**' Measure to Rows
- Drag '**Churn**' to Color under Marks
- Drag the '**CNT(combined)**' Dimension to Label under Marks
- Click the dropdown under the Churn Label > select 'Measure' > select 'Count'
- Drag the '**Datasource**' Dimension to the Filters column > click 'All' > click 'OK'
- Double click Tooltip under the Marks column and edit the Tooltips as appropriate
- Double click the title and edit
- Click the 'Churn' filter box and select 'Edit Colors...' from the dropdown > select the Color Blind color palette > make 'No' the lightest grey shade and 'Yes' the darkest red shade > click 'OK'
- Double-click the axis label that says 'Count of combined' > change the Axis Title to 'Customer Count' and then click 'X' to close this dialog box

Sheet 10: WGU Record Count (Text Table)

- Click the 'New Worksheet' tab > change the name of the sheet
- Drag the '**Datasource**' Dimension to Rows
- Drag '**CNT(combined)**' Measure to Text under Marks
- Drag the '**Datasource**' Dimension to the Filters column > select only 'wgu' > click 'OK'
- Double click the title and edit

Dashboard 1: Customer Profile

- Click the 'New Dashboard' tab > change the name of the sheet to 'Customer Profile'
- Drag and drop the sheets created above into the blank dashboard workspace, adjusting size as needed for each visualization
- Change the filters to floating and delete the column these are held within
- Move these filters next to their corresponding graphs

- Under Objects > Text, add a title to the dashboard and customize the churned customers filter with a heading describing the filter

Sheet 11: Record Count (Text Table)

- Click the 'New Worksheet' tab > change the name of the sheet
- Drag the '**Datasource**' Dimension to Rows
- Drag '**CNT(combined)**' Measure to Text under Marks
- Drag the '**Datasource**' Dimension to the Filters column > select 'All' > click 'OK'
- Double click the title and edit

Sheet 12: Customer Churn (Text Table)

- Click the 'New Worksheet' tab > change the name of the sheet
- Drag the '**Churn**' Dimension to Rows
- Drag '**combined(Count)**' Measure to Text under Marks
- Drag the '**Datasource**' Dimension to the Filters column > select 'All' > click 'OK'
- Double click the title and edit

Sheet 13: Churn Rate (KPI)

- Click the 'New Worksheet' tab > change the name of the sheet
- Under the Analysis tab > select 'Create Calculated Field..' > name this **Churn Rate** > enter the formula $\text{SUM}([\text{Churned Customers}]) / \text{COUNT}([\text{Churn}]) * 100$ > click 'OK'
- Drag this new 'Churn Rate' Measure to Text under the Marks Column. Click the drop down on this Measure > select Format... > under Numbers select Number (Custom) > Add a suffix of '%'
- Drag the '**Datasource**' Dimension to the Filters column > select 'All' > click 'OK'
- Double click the title and edit to remove (we'll add this later in the dashboard)

Sheet 14: Customer Lifetime Value by Contract Type (Highlight Table)

- Click the 'New Worksheet' tab > change the name of the sheet
- Drag the '**Contract Id**' Dimension to Rows
- Under the Analysis tab > select 'Create Calculated Field..' > name this **CLV** > enter the formula $[\text{Monthly Charge}] * [\text{Tenure}]$ > click 'OK'
- Drag this new 'CLV' Measure to Text under the Marks Column. Click the drop down on this Measure > select Measure > select Average. Click this drop down again and select Format... > under Numbers select Number (Custom) > Add a Prefix of '\$'
- Drag '**CLV**' Measure to Color under Marks. Click the drop down on this Measure > select Measure > select Average.
- Under Show Me, select the Highlight Table
- Drag the '**Datasource**' Dimension to the Filters column > select 'All' > click 'OK'
- Under the filter, adjust the title and format the color to a stepped greyscale
- Double click the title and edit

Sheet 15: Customer Lifetime Value by Partner & Dependent Status (Highlight Table)

- Click the 'New Worksheet' tab > change the name of the sheet

- Drag the '**Partner**' Dimension to Columns
- Drag the '**Dependents**' Dimension to Rows
- Drag '**CLV**' Measure to Text under Marks. Click the dropdown on this Measure > select Measure > select Average.
- Drag '**CLV**' Measure to Color under Marks. Click the dropdown on this Measure > select Measure > select Average.
- Under Show Me, select the Highlight Table
- Drag the '**Datasource**' Dimension to the Filters column > select 'All' > click 'OK'
- Under the filter, adjust the title and format the color to a stepped greyscale
- Double click the title and edit

Sheet 16: Contract Type By Partnership Status (Side-by-Side Bar Chart)

- Click the 'New Worksheet' tab > change the name of the sheet
- Drag the '**Partner**' & '**Contract Id**' Dimensions to Columns
- Drag the '**Partner**' Dimension to Rows > click Dropdown > select Measure > Count
- Drag '**Churn Rate**' Measure to Text under Marks. Select the dropdown > Format > customize the numbers to only show 1 decimal place of precision and add a % suffix
- Drag '**Partner**' Measure to Color under Marks.
- Under Show Me, select the Side-by-side bars
- Drag the '**Datasource**' Dimension to the Filters column > select 'All' > click 'OK'
- Under the filter, adjust the title and format the color the colorblind palette, selecting the lightest orange color for the 'No' item and darkest orange color for the 'Yes' item.
- Double click the title and edit

Sheet 17: Contract Type By Dependents (Side-by-Side Bar Chart)

- Click the 'New Worksheet' tab > change the name of the sheet
- Drag the '**Dependents**' & '**Contract Id**' Dimensions to Columns
- Drag the '**Dependents**' Dimension to Rows > click Dropdown > select Measure > Count
- Drag '**Churn Rate**' Measure to Text under Marks. Select the dropdown > Format > customize the numbers to only show 1 decimal place of precision and add a % suffix
- Drag '**Dependents**' Measure to Color under Marks.
- Under Show Me, select the Side-by-side bars
- Drag the '**Datasource**' Dimension to the Filters column > select 'All' > click 'OK'
- Under the filter, adjust the title and format the color the colorblind palette, selecting the lightest orange color for the 'No' item and darkest orange color for the 'Yes' item.
- Double click the title and edit

Sheet 18: Churn Rate by Contract Type (Treemap)

- Click the 'New Worksheet' tab > change the name of the sheet
- Drag the '**Contract Id**' Dimension to Text in the Marks column
- Drag the '**Churn Rate**' Measure to Color under Marks
- Drag '**Churn Rate**' Measure to Size under Marks
- Select Show Me, select Treemap
- Drag the '**Datasource**' Dimension to the Filters column > select 'All' > click 'OK'

- Under the filter, adjust the title and format the color a stepped greyscale palette
- Double click the title and edit

Sheet 19: Customer Churn by Tenure Length (Bar Chart with Average Line)

- Click the 'New Worksheet' tab > change the name of the sheet
- Drag the '**Tenure (bins)**' Dimension to Columns
- Drag the '**combined(Count)**' Measure to Rows
- Drag the '**Churn**' Dimension to Color under Marks. Click the drop down > select Measure > select Count.
- Drag '**Churned Customers**' Measure to Text under marks > select Measure > Sum
- Click Analytics > select Average Line
- Drag the '**Datasource**' Dimension to the Filters column > select 'All' > click 'OK'
- Under the filter, adjust the title and format the color a stepped warm greyscale palette
- Double click the title and edit

Dashboard 2: Competitor Analysis

- Click the 'New Dashboard' tab > change the name of the sheet to 'Competitive Analysis'
- Drag and drop the sheets created above into the blank dashboard workspace, adjusting size as needed for each visualization
- Add appropriate filters for each visual
- Customize filters by selecting drop down > Single Value (list) for Datasource, Multiple Values (dropdown) for Contract Id, Multiple Values (list) for Tenure (months).
- Under Objects > Text, add a title to the dashboard, the Data Filters column, and the Churn Rate KPI.

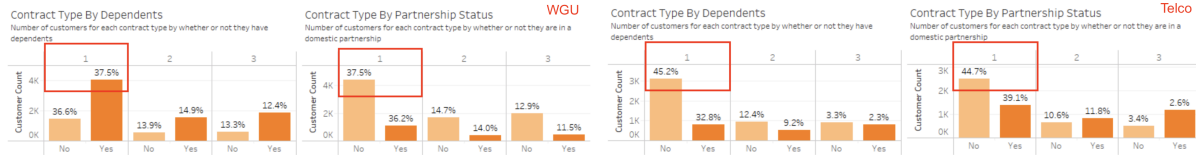
C5. Data Analysis Results

The supplemental competitor dataset's variables were carefully chosen due to their similarity to our own data, providing valuable insights into our own customer base. By comparing our data with the competitor data, we can gain a deeper understanding of customer demographics and churn factors.

1. Churn Rate Similarities Across Competitors: Both WGU Telecom and our competitor's customers exhibit nearly identical churn rates (~26.5%). This suggests that customer churn challenges are shared across the industry.



2. Month-to-Month (1) Contracts Have the Highest Churn: Across both datasets, customers with month-to-month contracts (represented by a Contract ID of 1) show the highest churn rate. This highlights that flexible, short-term contracts are more prone to churn, reinforcing the importance of encouraging long-term commitments.



3. Higher CLV for Longer-Term Contracts: In both datasets, customers on Two-Year Contracts (represented by a Contract ID of 3) have the highest Customer Lifetime Value (CLV). WGU Telecom customers on these contracts show a CLV of \$6,144.30, while our competitors' customers have a lower CLV of \$3,706.76, indicating our company's ability to extract more value from long-term contracts.

Customer Lifetime Value
Average CLV by Contract Type

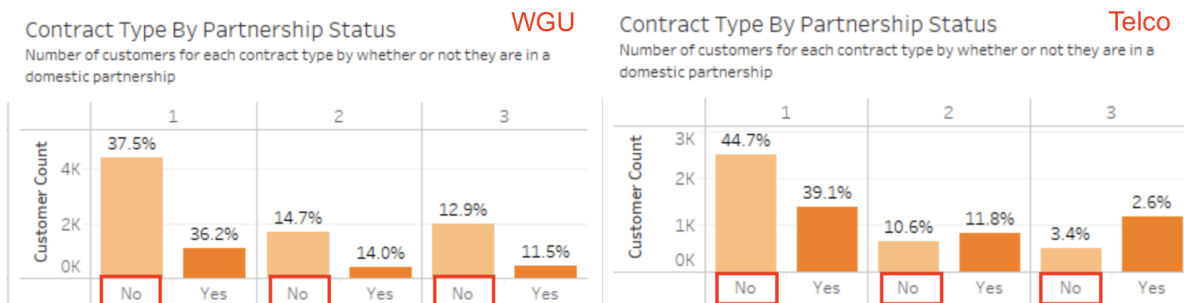
Contract Id		
3		\$6,144.30
2	WGU	\$5,924.32
1		\$5,884.45

Customer Lifetime Value
Average CLV by Contract Type

Contract Id		
3		\$3,706.76
2	Telco	\$3,029.83
1		\$1,370.12

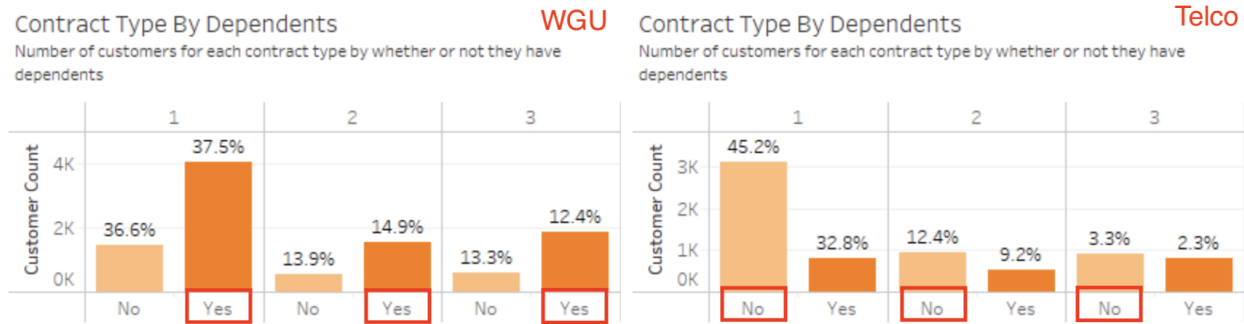
4. Partnership Status Affects Churn Similarly:

Across both datasets, customers who are not in a domestic partnership are more likely to churn. This could indicate that individuals who are not in long-term partnerships might place less value on stability and commitment in their service contracts. They may be more inclined to switch providers for better offers or due to dissatisfaction.



5. Dependent Status Affects Churn Differently:

For WGU Telecom, customers with dependents are more likely to churn, while interestingly in Telco's dataset, it's the customers without dependents who churn more. This divergence suggests different factors influencing family-related customer segments for each company. Our customers may be churning at a higher rate due to factors such as perceived service costs, unmet expectations for family-oriented services, or dissatisfaction with the service's value for families. Families tend to have higher service demands (higher data usage or more lines) which, if not met, could certainly lead to dissatisfaction and churn.



C6. Limitations of the Analysis

The analysis provides valuable insights but is constrained by several limitations. The assumption of complete similarity between WGU Telecom and the competitor, Telco, may not be entirely accurate. Factors such as branding, customer service differences, and market positioning could influence customer behavior in ways that the analysis does not account for. Our analysis is also heavily reliant on quantitative data, overlooking qualitative aspects such as customer feedback and intentions, which are important for understanding the underlying reasons for churn.

The use of SQL and Tableau, while effective for basic data analysis and visualization, also introduces challenges. These tools are well-suited for ad hoc analysis but lack the advanced statistical capabilities needed for hypothesis testing, regression modeling, and machine learning. This limits the ability to derive deeper insights and make accurate predictions. SQL's challenges with handling complex data transformations can also make preprocessing time-intensive and prone to errors.

Tableau's visualization capabilities, although great for visualizations directed towards general stakeholders, have limitations when it comes to dynamic and exploratory analyses. Additionally, both tools struggle with scalability and performance when analyzing large datasets or conducting more intricate analytical tasks. The focus on descriptive analytics further restricts the ability to conduct predictive or prescriptive analysis. Collaboration and reproducibility are also greatly hindered by the lack of version control and reliance on static queries and dashboards, especially given the direct connection to the database.

To address these limitations, future analyses should incorporate more granular data, qualitative research, and consider the impact of external factors. Adopting other tools like Python or R would enable us to perform more advanced statistical modeling, streamline our data processing processes, and allow us to create more enhanced visualizations. Combining these approaches can significantly improve the depth, accuracy, and actionable insights of the analysis.

D. Data & Code Sources

Blastchar. (2017). Telco Customer Churn. [Dataset]. *Kaggle*.

Retrieved December 1, 2024, from
<https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

Western Governors University. (2023). Churn Tables. *D211: Advanced Data Acquisition*. [Dataset].

Retrieved December 1, 2024, from
<https://access.wgu.edu/ASP3/aap/content/g9rke9s0rlc9ejd92md0.html>

E. References

Barron, S. (2023, July 13). How To Calculate Churn Rate in 5 Easy Steps [Definition + Formula]. *HubSpot*. [Blog Article].

Retrieved November 22, 2024, 2024, from
<https://blog.hubspot.com/service/what-is-churn-rate>

Bergman, H. (2024). hrbergman_d210_reflection_paper. *Western Governors University*. [Unpublished Research Paper].

Blastchar. (2017). Telco Customer Churn. [Dataset]. *Kaggle*.

Retrieved December 1, 2024, from
<https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

Kilin, I. (2022, April 27). The Best Charts for Color Blind Viewers. *Dataviz Best Practices*. [Datylon Blog].

Retrieved November 22 2024, 2024, from
<https://www.datylon.com/blog/data-visualization-for-colorblind-readers>

Macko, S. (2019, July 11). Telco Customer Churn (11.1.3+). *IBM Community: Business Analytics*. [IBM Community Resource].

Retrieved November 16, 2024, 2024, from
<https://community.ibm.com/community/user/businessanalytics/blogs/steven-macko/2019-07/11/telco-customer-churn-1113>

Nielson, J.. (2004, September 26). Checkboxes vs. Radio Buttons. [Blog Article].

Retrieved November 22, 2024, 2024, from
<https://www.nngroup.com/articles/checkboxes-vs-radio-buttons>

Shaffer, J. (2016, April 20). 5 Tips on Designing Colorblind-Friendly Visualizations. *Tableau*. [Blog Article].

Retrieved November 22, 2024, from
<https://www.tableau.com/blog/examining-data-viz-rules-dont-use-red-green-together>

Western Governors University. (2023). Churn Tables. *D211: Advanced Data Acquisition*. [Dataset].

Retrieved December 1, 2024, from
<https://access.wgu.edu/ASP3/aap/content/g9rke9s0rlc9ejd92md0.html>